

Nominal & Ordinal Scales

Mean, Median & Mode

INSTANT

Data Analysis Diploma • Session 03

DESCRIPTIVE STATISTICS

Types of Data · Measures of Central Tendency · Histogram & KDE · Range,
Variance & Standard Deviation · IQR & Box Plot

7 Goals**Core Outcomes**

Variables • Center • Spread

35 Slides**Curriculum Depth**

Distributions & Case Studies

Visual Labs**Hands-on Plots**

Histogram • KDE • Box Plot

Start Session 03 →

Variance & Std Dev

Box Plot & Outliers

— COMPETENCIES

What You Will Be Able To Do

Master fundamental statistical tools to summarize, visualize, and extract real insights from data:



Types of Data

Recognize categorical and numerical variables



Classify Variables

Differentiate nominal, ordinal, discrete, continuous



Measures of Central Tendency

Apply mean, median and mode correctly



Measure Variability

Quantify spread with range, variance, SD, IQR



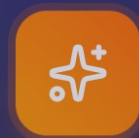
Read Visualizations

Interpret histograms, KDE plots and box plots



Detect Outliers

Spot extreme values and judge their impact



PART 2

Why Statistics Matters

How does a single number help a business make a confident decision?

THE SCIENCE OF DATA

What Is Statistics?

The bridge connecting raw tabular records to strategic business intelligence.

The science of collecting, organizing & interpreting data

- Descriptive statistics summarize what already happened in the data
- Inferential statistics use a sample to draw conclusions about a larger population
- Every dashboard metric — average order value, churn rate, satisfaction score — is a statistic
- Good statistics turn raw rows of data into a decision a manager can act on



Raw Data

Unprocessed table records

01



Organize & Summarize

Grouping, center & spread

02



Patterns Emerge

Distributions & shapes

03



Confident Decision

Actionable business strategy

04

— REAL-WORLD IMPACT

Statistics Powers **Everyday Business Decisions**

Modern enterprises make millions in revenue through statistical measurement:



Customer Spending

Average basket size guides pricing & promotions



Sales Analysis

Monthly trends reveal seasonality & growth



Employee Salaries

Pay equity checks across roles & departments



Student Performance

Score distributions flag who needs support



Product Ratings

Review spread separates loved vs polarizing products

METHODOLOGY COMPARISON

Descriptive vs Inferential Statistics

Understanding the scope and purpose of the two major disciplines in statistical analysis.

Descriptive Statistics

Summarize & Describe Existing Data

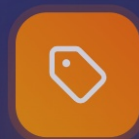
- Summarizes the data you already have
- Answers: what happened, on average, and how spread out is it?
- Tools: mean, median, mode, range, standard deviation, charts
- Example: "Our average delivery time last month was 3.2 days"

Inferential Statistics

Deduce & Predict Population Trends

- Uses a sample to draw conclusions about a larger population
- Answers: what can we expect going forward, with what confidence?
- Tools: hypothesis testing, confidence intervals, regression
- Example: "Based on 500 surveyed customers, we estimate 68% of all customers prefer next-day delivery"

💡 This session focuses entirely on descriptive statistics — inferential methods are covered in a later session.



PART 3

Types of Data

Every statistical method depends first on knowing what kind of data you are looking at.

— CATEGORICAL MEASURES

Qualitative (Categorical) Data

Describes qualities or categories — cannot be meaningfully measured on a numeric scale.

Nominal

No Inherent Order

- Categories with no natural order or hierarchy
- Example: Department — Sales, IT, Marketing, HR
- Example: Payment Method — Cash, Card, Wallet
- Cannot be ranked, added, or averaged

Ordinal

Meaningful Ranking

- Categories with a meaningful order or sequence
- Example: Performance Rating — Low, Medium, High
- Example: Customer Satisfaction — Poor to Excellent
- Order matters, but exact gaps between levels are not defined

— NUMERICAL MEASURES

Quantitative (Numerical) Data

Expressed as numbers that can be measured, ordered and used in calculations.

Discrete

Countable Whole Numbers

- Countable, whole-number values (integers only)
- Example: Number of support tickets per day
- Example: Number of children in a household
- Cannot take fractional values in real-world practice

Continuous

Measurable Infinite Continuum

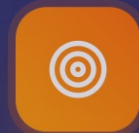
- Measurable values on an infinite, unbroken scale
- Example: Employee salary or delivery time in minutes
- Example: Product weight or environmental temperature
- Can take any value within a range, including decimals

— STEVENS TYPOLOGY

The Four Measurement Scales

Categorization framework determining which statistical formulas are mathematically valid.

Scale	Definition	Example	Math Operations
Nominal	Categories, no order	Department, Gender	Count, Mode only
Ordinal	Categories, ordered	Rating: Low / Medium / High	Count, Mode, Median, ranking
Interval	Ordered, equal gaps, no true zero	Temperature (°C), Calendar year	+ and – meaningful; ratios are not
Ratio	Ordered, equal gaps, true zero	Salary, Age, Sales revenue	All operations, including ratios



PART 4

Measures of Central Tendency

"How can one single number summarize thousands of observations?"

BALANCING POINT

Mean — The Balancing Point

The exact center of gravity where the sum of positive deviations balances negative deviations.

DEFINITION & FORMULA

$$\text{Mean } (\bar{x}) = (\Sigma \text{ values}) \div n = 87 \div 5 = 17.40$$

- Sum of all values divided by the count of values
- Mean = $(\Sigma \text{ values}) \div n$
- Business example: average order value across 5 transactions of \$12, \$15, \$18, \$20, \$22 is \$17.40
- Advantage: uses every data point, mathematically well understood
- Limitation: pulled strongly toward extreme values (outliers)

Fulcrum Physics: Sum of Distances = 0



Negative deviations (-5.4, -2.4) cancel positive deviations (+0.6, +2.6, +4.6)
= 0!

— ROBUST CENTER

Median — The Middle Value

The 50th percentile dividing ordered data exactly into lower and upper halves.

DEFINITION & FORMULA

Odd: $(n + 1)/2 = 3\text{rd item}$ | **Even:** Avg of two centers

- The middle value once data is sorted from lowest to highest
- Odd count: the exact middle observation: $(n + 1) / 2$
- Even count: arithmetic average of the two middle observations
- Business example: median household income is preferred over mean because it resists billionaire outliers
- Handles skewed distributions and extreme outliers far better than the mean

Sorted Dataset (Lowest to Highest)



Median = 18

Exactly 2 observations below, and 2 observations above.

CATEGORICAL CENTER

Mode — The Most Frequent Value

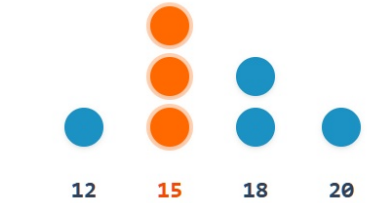
The highest frequency peak in discrete or categorical observations.

DEFINITION & APPLICATIONS

Frequency Peak = Most repeated observation

- The value (or category) that occurs most often in the dataset
- The ONLY measure of center that works on categorical/nominal data
- A dataset can be unimodal, bimodal, multimodal, or have no mode
- Business example: the most frequently ordered shoe size or menu item guides inventory planning
- Business example: most common customer complaint category directs process improvement

Frequency Dot Plot



Mode = 15 (appears 3 times)

Works on discrete numbers AND text labels.

COMPARISON TABLE

Mean vs Median vs Mode

Choosing the optimal measure of central tendency based on distribution shape.

Measure	Definition	Advantages	Best Used When
Mean	Arithmetic average of all values	Uses all data, well understood	Data is roughly symmetric, no extreme outliers
Median	Middle value in sorted order	Resistant to outliers and skew	Data is skewed or contains extreme values
Mode	Most frequently occurring value	Works with categorical data	You need the most common category or value

SENSITIVITY ANALYSIS

The Effect of Outliers

Demonstrating how a single extreme value pulls the mean while the median stands firm.

Interactive Outlier Simulator:

Remove Outlier (85) ↻

Original Dataset: [12, 14, 15, 16, 17, 18, 20]

Mean: 16.4 Median: 16



With Outlier (85 Added):

Mean: 24.1 (+47% skewed!) Median: 16 (0% shift – resistant!)





PART 5

Measures of Spread

"Can two datasets share the same average but behave completely differently?"

— MAX MINUS MIN

Range — The Simplest Measure

The total span between the absolute minimum and maximum observed values.

• Calibrated Quantitative Scale | Ordered Dataset (n = 6)

Standard (Range = 10)

⚠ Simulate Outlier (44)

DEFINITION & FORMULA

MEASURE OF SPREAD

Formula:

$$\text{Range} = \text{Maximum} - \text{Minimum}$$

$$= 22 - 12$$

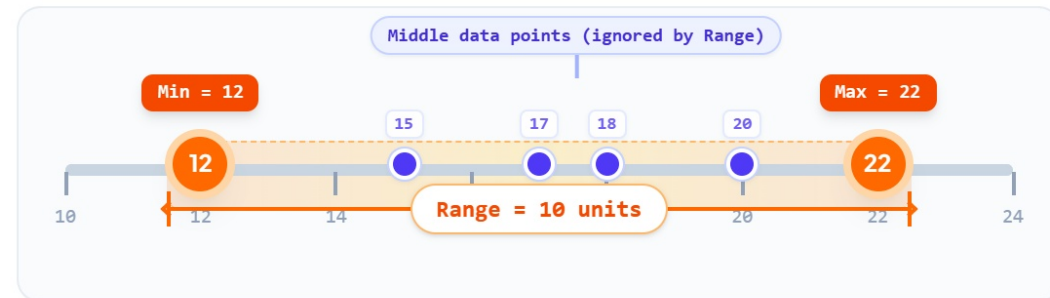
$$= 10 \text{ units}$$

- Range = Maximum value – Minimum value
- Business example: delivery times ranging from 12 to 22 hours gives a range of 10 hours
- Advantage: extremely quick to calculate and easy to explain to stakeholders
- Limitation: uses only two data points, completely ignoring the distribution in between

Visual Number Line Calibration

Accurate mathematical coordinates along the continuous axis

Span: 10 units



— SQUARED DEVIATIONS

Variance — Average Squared Distance

The average of squared differences from the mean, weighting larger gaps progressively.

CONCEPTUAL INTUITION

$$\text{Variance } (s^2) = \sum (x - \bar{x})^2 \div (n - 1)$$

- Measures how far, on average, each value sits from the mean
- Each distance is squared before averaging, so bigger gaps count exponentially more
- Squaring also makes every distance positive, so negative deviations do not cancel out
- A larger variance means the data is more spread out around the mean
- Expressed in squared units (e.g., dollars squared), which is why standard deviation is preferred

Squared Distance from Mean (17.4)

Point 12	Gap: -5.4	Squared: 29.16
Point 15	Gap: -2.4	Squared: 5.76
Point 18	Gap: +0.6	Squared: 0.36
Point 20	Gap: +2.6	Squared: 6.76
Point 22	Gap: +4.6	Squared: 21.16

Sum of Squares = 63.20 | Variance = 15.80

— ORIGINAL UNITS

Standard Deviation — Typical Distance From the Mean

The gold standard benchmark for measuring spread in the original variable units.

INTERPRETATION

$$s = \sqrt{\text{Variance}} = \sqrt{15.80} = 3.97$$

- The square root of variance — brought back into the original units of the variable
- Low standard deviation: observations cluster tightly around the mean
- High standard deviation: observations are spread widely, less predictable
- Business example: a delivery service with low SD in delivery time is far more reliable
- Widely used in Quality Control (Six Sigma) to flag defects and compare consistency

Distribution Width Comparison



Low Std Dev

Tightly clustered, consistent



High Std Dev

Widely dispersed, high variance

— RESISTANT SPREAD

Interquartile Range (IQR)

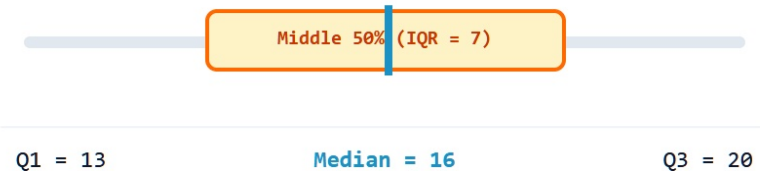
The spread of the middle 50% of sorted observations, completely resistant to outliers.

DEFINITION & FORMULA

$$\text{IQR} = Q3 \text{ (75\%)} - Q1 \text{ (25\%)} = 20 - 13 = 7.0$$

- Quartiles split sorted data into four equal parts: Q1 (25%), Median/Q2 (50%), Q3 (75%)
- $\text{IQR} = Q3 - Q1$ — the spread of the middle 50% of the data
- Not affected by extreme values or long tails at either end
- Business example: IQR of employee salaries shows the typical pay band, ignoring executive pay
- The standard mathematical basis for detecting outliers on a box plot

Middle 50% Coverage (Resistant to Outliers)

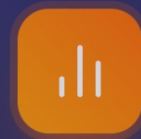


SELECTION CRITERIA

Choosing the Right Spread Measure

Comparative cheat sheet for selecting the appropriate variability metric.

Measure	What It Uses	Sensitive to Outliers?	Best Used When
Range	Max and min only	Very sensitive	A fast, rough sense of total spread is enough
Variance	Every value's squared distance from mean	Sensitive	A statistical basis for further calculations is needed
Standard Deviation	Square root of variance	Sensitive	Reporting typical spread in original, interpretable units
IQR	Middle 50% of sorted data (Q3 - Q1)	Resistant	Data has outliers or is skewed



PART 6

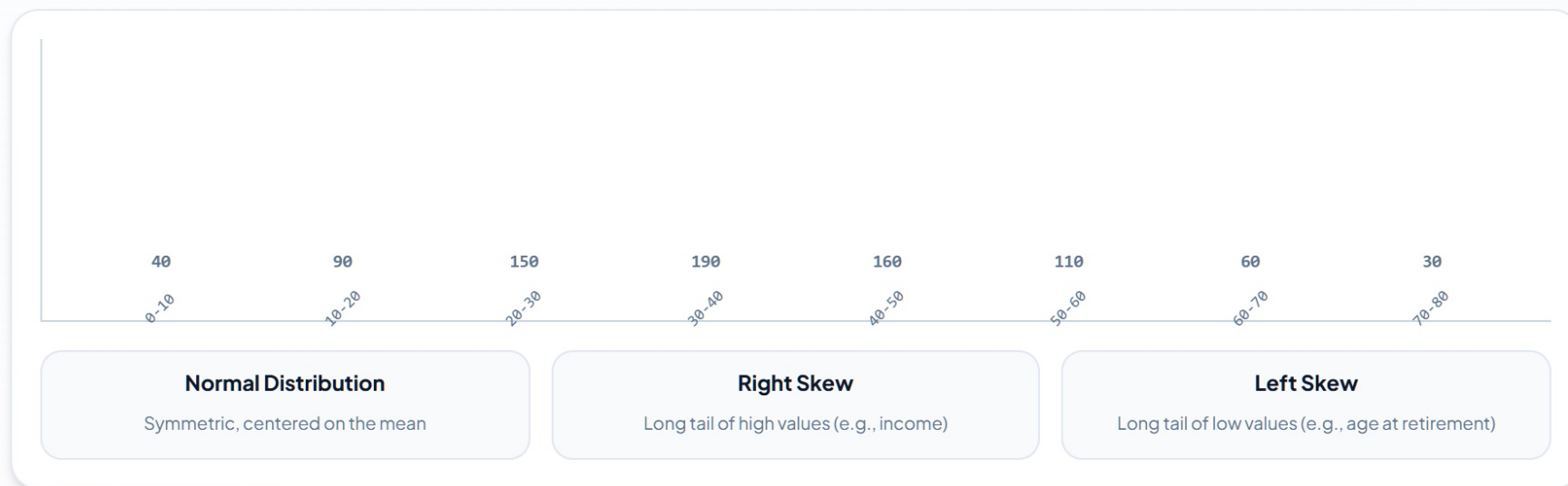
Statistical Visualizations

"A well-built chart can reveal in one glance what pages of numbers cannot."

— HISTOGRAM ANATOMY

Histogram — Frequency Across Bins

Data is grouped into equal-width bins; bar height shows how many observations fall in each bin.



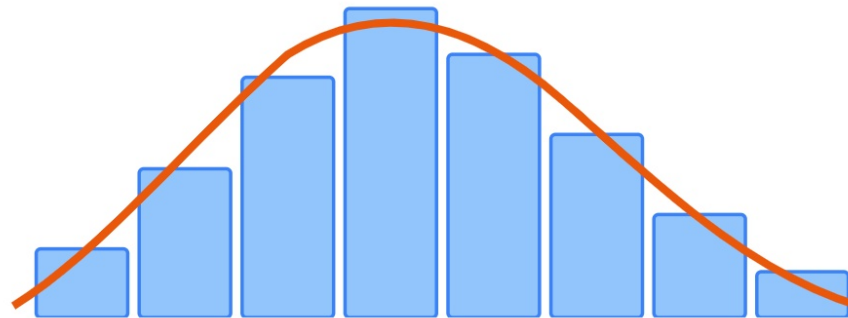
— CONTINUOUS CURVE

KDE Plot — A Smoothed View of Distribution

A Kernel Density Estimate (KDE) draws a smooth curve through the data, avoiding arbitrary bin-width choices.

■ Binned Histogram Counts

— KDE Smoothed Continuous Density



Avoids arbitrary bin boundaries by estimating the continuous probability density function.

BOX & WHISKER

Box Plot — Five Numbers, One Picture

The box spans Q1 to Q3 (the IQR), the line marks median, whiskers extend to min/max, and outliers are flagged.

Horizontal Box & Whisker Anatomy



MIN

10

Q1 (25%)

25

MEDIAN (50%)

40

Q3 (75%)

60

OUTLIER

92, 98

— INTERACTIVE PRACTICE

Reading Statistical Charts

Look at the shapes below and test your diagnostic skills on real distribution patterns:

01

Which dataset is most likely to contain outliers?

Click to reveal answer

02

Which distribution shows the greatest variability?

Click to reveal answer

03

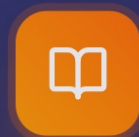
Which shape is skewed, and in which direction?

Click to reveal answer

04

Which measure of center would you trust most — mean or median?

Click to reveal answer



PART 7

Guided Practice

A realistic employee dataset — let's classify, calculate, and interpret together.

— TEN EMPLOYEE RECORDS

Employee Dataset

Representative workforce sample containing categorical, discrete, continuous, and outlier features.

Employee ID	Department	Salary (\$)	Age	Performance	Years Exp.
E001	Sales	\$52,000	29	4.2	3
E002	IT	\$68,000	34	3.8	6
E003	Marketing	\$55,000	27	4.5	2
E004	Sales	\$51,000	41	3.5	12
E005	IT	\$72,000	38	4.0	9
E006	HR	\$49,000	45	3.9	15
E007	Marketing	\$58,000	31	4.7	4
E008	IT	\$250,000	36	4.1	8
E009	Sales	\$53,000	26	3.6	1
E010	HR	\$47,000	50	3.7	20

— HANDS-ON TASKS

Calculate and Classify

Systematic 6-step walkthrough to extract full descriptive intelligence from the dataset:

Step 1

Identify Data Types

Identify the data type of each column in the dataset

Step 2

Measures of Central Tendency

Calculate the mean, median and mode of Salary

Step 3

Measures of Spread

Calculate the range, variance and standard deviation of Salary

Step 4

Calculate IQR

Calculate Q1, Q3 and the IQR of Salary

Step 5

Histogram Distribution

Sketch what the Salary histogram would look like — normal or skewed?

Step 6

Box Plot Outlier Check

Predict what a Salary box plot would reveal about E008

— OUTLIER DIAGNOSTIC

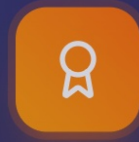
Interpreting the Salary Distribution

E008's salary sits well beyond the upper whisker — exactly the kind of point a box plot surfaces at a glance.

Salary Distribution (\$k) — Spotting Executive Outlier



MIN \$47,000	Q1 (25%) \$51,250	MEDIAN (50%) \$54,000	Q3 (75%) \$67,000	OUTLIER E008: \$250k
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PART 8

Summary

From raw numbers to confident, data-driven decisions.

— CORE PILLARS REVIEW

What We Covered Today

Key competencies mastered in Descriptive Statistics Part 1:



Types of Data

Nominal, ordinal, discrete, continuous



Measures of Central Tendency

Mean, median, mode



Measures of Spread

Range, variance, SD, IQR



Outliers

How extreme values distort the mean



Histogram

Frequency across bins, distribution shape



KDE Plot

A smoothed view of the distribution

— FORMULA REFERENCE

Quick Reference — Formulas at a Glance

Comprehensive mathematical formulas and their resistance to extreme values.

Measure	Formula	Resistant to Outliers?
Mean	$\Sigma \text{ values} \div n$	No
Median	Middle value of sorted data (avg. of two middles if n is even)	Yes
Mode	Most frequently occurring value	Yes
Range	Maximum - Minimum	No
Variance	Average of squared distances from the mean	No
Standard Deviation	Square root of variance	No
IQR	$Q3 - Q1$ (middle 50% of sorted data)	Yes



Think Statistically.

Every dataset tells a story — center, spread, and shape are how you learn to read it.

Next session: Descriptive Statistics Part 2

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